

Faik Kerem Ors

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EDUCATION

Purdue University

Ph.D. in Computer Science

GPA: 3.90/4.00, **Academic Advisor:** Prof. Elisa Bertino

Focus: Network Security

West Lafayette, IN, US

Aug. 2022 – May. 2027 (Anticipated)

Sabanci University

M.Sc. in Computer Science and Engineering

GPA: 3.87/4.00, **Thesis Advisor:** Prof. Albert Levi

Thesis: Data Driven Intrusion Detection for 6LoWPAN Based IoT Systems

Istanbul, Turkey

Sep. 2019 – Jan. 2022

Sabanci University

B.Sc. in Computer Science and Engineering

Minor in Mathematics

GPA: 3.86/4.00, **Ranked 3rd**

Istanbul, Turkey

Sep. 2014 – June 2019

RESEARCH INTERESTS

Network security, cellular network security, protocol reverse engineering, IoT security and privacy, systems and software security, machine learning for security.

RESEARCH EXPERIENCE

Research Intern

Mitsubishi Electric Research Laboratories (MERL)

May 2026 – Present

Cambridge, MA, USA

- Conducting research on fault and anomaly detection in large-scale multi-hop IoT networks.
- Designing and implementing fault and attack scenarios in a real IoT testbed to support data collection, evaluation, and model training.
- Developing network anomaly detection techniques leveraging large language models (LLMs) and graph neural networks (GNNs) to detect and analyze network failures and security threats.

Ph.D. Researcher

Purdue University - Computer Science

Aug. 2022 – Present

West Lafayette, IN, USA

- Developing methods to find vulnerabilities in cellular network security, specifically Open Radio Access Networks (O-RAN), using LLMs and formal verification tools.
- Designing and curating large-scale testbeds and datasets for vulnerability analysis and detection in O-RAN.
- Analyzing the security and privacy of communication protocols, particularly unknown or proprietary protocols, using deep learning techniques.
- Developing models and tools to systematically infer protocol structures and behaviors to enhance security analysis, anomaly detection, and formal verification.
- Contributed to the GSMA Open Telco Assets initiative with AT&T and academic partners, including O-RAN-grounded synthetic Q&A dataset generation and NDCG-based evaluation of telecom-domain embedding models for retrieval tasks.
- Advisor: Prof. Elisa Bertino

Research Assistant

Purdue University - Electrical and Computer Engineering

Aug. 2025 – May 2026

West Lafayette, IN, USA

- Conducted research on the theoretical foundations of TSD-based pavement evaluation, including deflection theory and pavement response under moving loads.
- Performed literature review, pavement modeling, and index development for TSD-based roadway condition evaluation methods.
- Designed and implemented Rust-based asynchronous clients for sensor data acquisition and device communication over TCP for Traffic Speed Deflectometer (TSD) systems.

- Integrated new sensor systems into a modular, real-time data pipeline using NATS JetStream and Arrow IPC for efficient streaming, serialization, and processing.
- Built frontend applications and APIs to visualize and manage roadway sensor data for the Indiana Department of Transportation (INDOT).
- Presented this work at the *2026 Purdue Road School Transportation Conference and Expo* and the *Joint Transportation Research Program (JTRP)*.
- Supervisors: Prof. James V. Krogmeier, Andrew Balmos

CyberPowder Fellow (NSF CyberTraining Award #2417934)

Jan. 2026 – May 2026

University of Utah - POWDER Wireless Testbed

Salt Lake City, UT, USA (Remote + On-site)

- Selected fellow in an NSF-funded wireless networking training and research mentoring program for US-based graduate students.
- Completed weekly research-focused training on wireless communications and hands-on experimentation on POWDER, the world's largest openly-accessible software-defined wireless testbed.
- Participated in a week-long on-site research exploration at the University of Utah, executing an experimental research plan on O-RAN vulnerability validation on the POWDER platform.

Research Assistant – Technical Team Lead

June 2025 – Aug. 2025

Purdue University - OATS, IoT4Ag, Agricultural and Biological Engineering

West Lafayette, IN, USA

- Installed and configured LoRaWAN-based soil and weather sensors, including the deployment of various LoRaWAN gateways as part of the SPRING (Solar-Powered Remote IoT4Ag Network Gateway) inventory.
- Implemented communication interfaces, sensor data decoding logic in Chirpstack, RedPanda Connect flows, TimescaleDB tables, and Grafana dashboards for data pipelining and visualization.
- Built a public-facing dashboard to visualize real-time sensor data and deliver interactive learning content for students, teachers, and the public.
- Collaborated in weekly meetings, shared technical design decisions, and co-developed lecture material, visuals and tutorials that demystify IoT systems for middle-school students.

Research Intern

June 2020 – June 2021

Purdue University - GoBoiler Internship Program (Selected Attendee)

West Lafayette, IN, USA

- Implemented a secure and robust context-based group pairing scheme for heterogeneous IoT devices.
- This work resulted in a publication at *IEEE Symposium on Security and Privacy (S&P 2023)*.
- Supervisor: Dr. Z. Berkay Celik

Summer Research Intern

June 2018 – Sep. 2018

Technical University of Berlin

Berlin, Germany

- Implemented deep learning models to optimize the bitrate selection decision on DASH clients.
- Reviewed the literature and delivered an overview presentation on Dynamic Streaming over HTTP (DASH).
- Attended M.Sc. lectures and seminars with the focus on content delivery techniques.
- Supervisors: Dr. Suzan Bayhan and Prof. Abdel-Karim Al-Tamimi

PEER-REVIEWED PUBLICATIONS

- **Faik Kerem Ors**, Rumeysa Gorgulu, Andrew D. Balmos, Fabio A. Castiblanco, Joshua K. Bailey, Kristin Korcha, Kwabena Bayity, James V. Krogmeier, and Yaguang Zhang. SPRING Portal: A Multi-Site IoT Data Platform for Classroom-Ready Agricultural Education. In *ASABE AIM 2026*, Indianapolis, IN, July 2026. (Accepted)
- **Faik Kerem Ors**, Andrew D. Balmos, Rumeysa Gorgulu, Sofia Gerena Godoy, James V. Krogmeier, and Yaguang Zhang. Agriculture-Focused Introduction to Data Visualization with Grafana: A University-Level Tutorial Module. In *ASABE AIM 2026*, Indianapolis, IN, July 2026. (Accepted)
- Rumeysa Gorgulu, **Faik Kerem Ors**, Kristin Korcha, and Yaguang Zhang. From SEED to SPRING: A Sensor-Based Agriculture Module for Middle School Education. In *ASABE AIM 2026*, Indianapolis, IN, July 2026. (Accepted)
- Habiba Farrukh*, Muslum Ozgur Ozmen*, **Faik Kerem Ors**, Z. Berkay Celik. One Key to Rule Them All: Secure Group Pairing for Heterogeneous IoT Devices. In *IEEE Security and Privacy (S&P 2023)*. – **Cited by 32**
- **Faik Kerem Ors**, and Albert Levi. Data driven intrusion detection for 6LoWPAN based IoT systems. In *Ad Hoc Networks*, 143, pages 103-120, April 2023. – **Cited by 15**

- **Faik Kerem Ors.** Data Driven Intrusion Detection for 6LoWPAN Based IoT Systems. *M.Sc. Thesis*, December 2021.
- **Faik Kerem Ors**, Mustafa Aydin, Aysu Bogatarkan, and Albert Levi. Scalable Wi-Fi Intrusion Detection for IoT Systems. In *11th IFIP International Conference on New Technologies, Mobility and Security (Security Track)*, Paris, France, April 2021. – **Cited by 9**
- **Faik Kerem Ors**, Suveyda Yeniterzi, and Reyyan Yeniterzi. Event Clustering within News Articles, In *Proceedings of the Workshop on Automated Extraction of Socio-political Events from News 2020*, pages 63–68, Marseille, France, May 2020. European Language Resources Association (ELRA). (Proposed system ranked 1st in the shared task). – **Cited by 29**

CONFERENCE PRESENTATIONS

- Pavement Monitoring: Integrating Edge Computing and High-Speed Deflectometry. In *2026 Purdue Road School Transportation Conference and Expo*, West Lafayette, IN, March 2026.
- Scalable Wi-Fi Intrusion Detection for IoT Systems. In *11th IFIP International Conference on New Technologies, Mobility and Security (Security Track)*, Paris, France, April 2021.
- Event Clustering within News Articles, In *Proceedings of the Workshop on Automated Extraction of Socio-political Events from News 2020*, pages 63–68, Marseille, France, May 2020. European Language Resources Association (ELRA).

POSTER PRESENTATIONS

- *Development of a Traffic Speed Deflectometer for Indiana.* (Poster) Joint Transportation Research Program (JTRP), Indiana Government Center North, Indianapolis, IN, March 4, 2026.

TEACHING EXPERIENCE

- Invited Lecturer** Fall 2026
ASM 591: Ag Data Visualization & Edge Computing, Purdue University *West Lafayette, IN, US*
 • Delivered a lecture and lab, and assigned coursework on database management systems, SQL, and Grafana for agricultural and weather data visualization.
- Teaching Assistant** Aug. 2022 – May 2025
Purdue University *West Lafayette, IN, US*
 • Held office hours, supervised student projects, designed and graded exams and assignments.
 • Courses (reverse chronological): Computer Security (CS 426; Spring 2025, Fall 2023, Spring 2023), Cryptography (CS 555; Fall 2024), Information Security (CS 526; Spring 2024), Security Analytics (CS 529; Fall 2022)
- Guest Lecturer** Fall 2023
CS 426: Computer Security, Purdue University *West Lafayette, IN, US*
 • Delivered lectures on buffer overflows, return oriented programming, SQL injection, and cross-site scripting.
- Teaching Assistant** Feb. 2018 – Jan. 2022
Sabancı University *Istanbul, Turkey*
 • Held office hours, designed and graded assignments, conducted lab sessions and supervised student projects.
 • Courses (reverse chronological): Computer Networks (CS 408; 2022, 2021, 2020), Computer and Network Security (CS 432), Machine Learning (CS 412), Advanced Programming (CS 204), Database Systems (CS 306)

SERVICES

- Reviewer in IEEE Transactions on Information Forensics and Security (TIFS) 2026, 2025, 2024
- External Reviewer in NDSS 2026, 2025, 2024, 2023
- External Reviewer in IEEE S&P 2026, 2025, 2024
- External Reviewer in USENIX Security 2026, 2024, 2023
- Reviewer in ITU Journal on Future and Evolving Technologies (ITU J-FET) 2022
- Reviewer in IEEE International Conference on Communications (ICC) 2022
- Reviewer in IEEE Conference on Communications and Network Security (CNS) 2021
- Reviewer in The Computer Journal (Oxford University Press) 2021, 2020

HONORS AND AWARDS

- Selected as a CyberPowder Fellow, an NSF-funded wireless networking research program (December 2025).
- Graduate Teaching Award in Recognition of the Teaching Performance during Spring 2025 (November 2025).
- Tuition waiver and Graduate Teaching Assistantship offer by Purdue University for graduate studies (2022 – 2027).
- Full tuition waiver and stipend by Sabancı University for graduate studies (2019 – 2021).
- Dean’s High Honor List, Sabancı University (2016 – 2019).
- Recipient of Sakıp Sabancı Encouragement Scholarship, which covers 100% of tuition fee, because of academic excellence (2016 – 2019).

TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#, Java, Rust, SQL

Frameworks and Libraries: PyTorch, Tensorflow, Pandas, NumPy, Scikit-learn, Keras, Flask, Django

Operating Systems: Unix, Linux, macOS, Windows

Technologies: git, MySQL, PostgreSQL, Docker, JUnit, Android Studio, Chirpstack, Redpanda Connect, Grafana

Tools: Wireshark, Metasploit, Hashcat, Burp Suite, Nmap, SQLmap, Wfuzz, IDA, Binwalk, The Harvester, Dirbuster

ADDITIONAL WORK EXPERIENCE

R&D Engineer

Feb. 2019 – Aug. 2019

PRODAFT

Istanbul, Turkey

- Implemented a machine learning based phishing detection system from scratch.
- Developed RESTful microservices to be integrated into the threat intelligence ecosystem of the company.

Security Research Intern

July 2017 – Sep. 2017

PRODAFT

Istanbul, Turkey

- Worked on penetration testing and developed penetration testing tools in Python.

OUTREACH

- Contributed to a hands-on outreach session on AI and cyber-physical systems for Indiana 4-H summer camp students at Purdue (August 2024), helping demonstrate how these technologies apply to real-world challenges.